Supplementary Information
On The Conservation Laws and Pathway Realizability of Biochemical Reaction Networks

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1 Software architecture and analysis interfaces

This supplement describes the chemical reaction network (CRN) submodule of `SynKit`. Figure S1 shows its input adapters and structural analyses built on the shared reaction-system representation. The supplementary examples specify their reaction equations and initial inventories, allowing their results to be interpreted separately from the reduced glycolysis model in the main text.

The numerical results and timings were regenerated on 6 November 2026 with `SynKit` 1.6.5 (commit `5951b401b161`) by `analysis/scripts/regenerate_evidence.py`, as in the main text. Source hashes, native build identifiers, and input checksums accompany the results. The external `CRNT4SBML` comparison uses its recorded outputs on checksum-identical exported inputs.

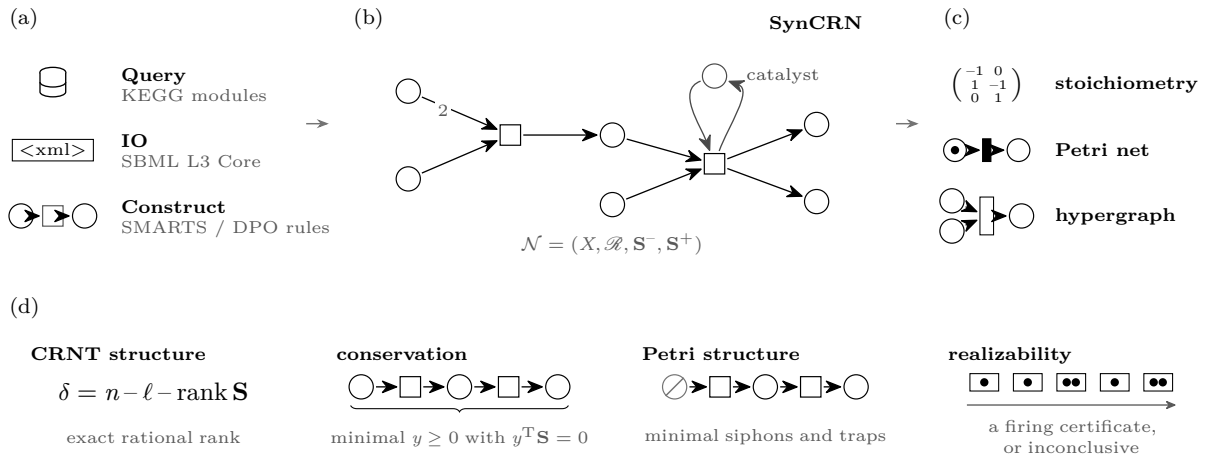


Fig. S1: Software architecture and data flow in the CRN submodule of `SynKit`. **(a)** Input adapters for `KEGG` pathways, `SBML` Core, and reaction rules. **(b)** Shared core records preserving reactant and product multiplicities. **(c)** Derived stoichiometric, Petri-net, and hypergraph representations. **(d)** Structural analyses: exact rational CRNT invariants, minimal conserved pools, Petri-net siphons/traps, and certified discrete pathway realizability (state/depth limits return `inconclusive`). Icons are schematic.

2 Reaction-network case studies

The two examples examine phosphoryl-donor availability in a listed glycolysis network and carbon-precursor multiplicity in a rule-generated formose network. The calculations use the equations and inventories reported below, and do not infer kinetic or physiological behavior.

2.1 Phosphoryl-donor requirements in an extended glycolysis network

This listed network combines upper glycolysis with cofactor-coupled alternatives. It is distinct from the 14-species, currency-reduced `KEGG` module used in the main text. Its purpose is to compare qualitative reachability with realizability from a finite ATP inventory.

The candidate routes share hexose phosphorylation and aldolase cleavage, but differ in whether the second phosphorylation consumes ATP or ADP. The following tables specify the reaction equations and putative species annotations.

2.1.1 Reaction stoichiometry and metabolite annotation

The `Structure` module constructs a `SynCRN` object from 34 directed reactions involving 40 species. Table S1 lists the complete equations, with identifiers $r1, \ldots, r34$. The biochemical names below are putative annotations, and the structural calculations do not independently validate these identities or enzyme assignments.

Table S1: Directed reaction stoichiometry for the 40-species, 34-reaction extended glycolysis network. Identifiers and coefficients correspond to species annotations in Table S2.

Rule	Reaction
r1	A + B → C + D
r2	E → B + F
r3	C + G → A + H
r4	I → J
r5	J + K + L → M + N + O
r6	J + K + P → M + O + Q
r7	H → J + I
r8	A + M → R + C
r9	R → E
r10	S + C → A + T
r11	S + U → T + V
r12	A + G → W + H
r13	J + 2X + F → 2O + R + 2Y

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Table S1 (continued from previous page)

Rule	Reaction
r14	A + S → T + W
r15	T → G
r16	Z + C → A + B + AA
r17	Z + AB → B + AA + AC
r18	B + F → E
r19	Z + AD → B + AA + AE
r20	F + H → K + G
r21	J → I
r22	M + N + O → J + K + L
r23	M + O + Q → J + K + P
r24	J + I → H
r25	R + C → A + M
r26	E → R
r27	AF + D → AA + AG
r28	L + AH + D → N + O + AA + AI
r29	P + AH + D → O + Q + AA + AI
r30	2X + AH + D → 2O + 2Y + AA + AI
r31	AH + AJ → AI + AK
r32	AG + AL → AF + AJ
r33	L + AK → N + O + AL
r34	AH + AM + D → AA + AI + AN

Species tokens are associated with putative biochemical names and SMILES in Table S2. Structural calculations use the listed reaction coefficients, while annotations aid interpretation and do not independently validate the chemistry.

Table S2: Species tokens, putative biochemical names, and SMILES strings for the extended glycolysis network (Table S1).

Token	Putative biochemical identity	SMILES
A	ADP	`Nc1ncnc2c1ncn2[C@@H]1O[C@H](COP(=O)(O)OP(=O)(O)O)[C@@H](O)[C@H]1O`
B	phosphoenolpyruvate	`C=C(OP(=O)(O)O)C(=O)O`
C	ATP	`Nc1ncnc2c1ncn2[C@@H]1O[C@H](COP(=O)(O)OP(=O)(O)OP(=O)(O)O)[C@@H](O)[C@H]1O`

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Table S2 (continued from previous page)

Token	Putative biochemical identity	SMILES
D	pyruvate	CC(=O)C(=O)O
E	2-phospho-D-glycerate	O=C(O)[C@@H](CO)OP(=O)(O)O
F	water	O
G	D-fructose-6-phosphate	O=P(O)(O)OC[C@H]1OC(O)(CO)[C@@H](O)[C@@H]1O
H	D-fructose-1,6-bisphosphate	O=P(O)(O)OC[C@H]1OC(O)(COP(=O)(O)O)[C@@H](O)[C@@H]1O
I	dihydroxyacetone phosphate	O=C(CO)COP(=O)(O)O
J	D-glyceraldehyde-3-phosphate	O=C[C@H](O)COP(=O)(O)O
K	orthophosphate	O=P(O)(O)O
L	NAD^+	NC(=O)c1ccc[n+]([C@@H]2O[C@H](COP(=O)(O)OP(=O)(O)OC[C@H]3O[C@@H](n4cnc5c(N)ncnc54)[C@H](O)[C@@H]3O)[C@@H](O)[C@H]2O)c1
M	1,3-bisphospho-D-glycerate	O=C(OP(=O)(O)O)[C@H](O)COP(=O)(O)O
N	NADH	NC(=O)C1=CN([C@@H]2O[C@H](COP(=O)(O)OP(=O)(O)OC[C@H]3O[C@@H](n4cnc5c(N)ncnc54)[C@H](O)[C@@H]3O)[C@@H](O)[C@H]2O)C=CC1
O	proton	[H+]
P	$NADP^+$	NC(=O)c1ccc[n+]([C@@H]2O[C@H](COP(=O)(O)OP(=O)(O)OC[C@H]3O[C@@H](n4cnc5c(N)ncnc54)[C@H](OP(=O)(O)O)[C@@H]3O)[C@@H](O)[C@H]2O)c1
Q	NADPH	NC(=O)C1=CN([C@@H]2O[C@H](COP(=O)(O)OP(=O)(O)OC[C@H]3O[C@@H](n4cnc5c(N)ncnc54)[C@H](OP(=O)(O)O)[C@@H]3O)[C@@H](O)[C@H]2O)C=CC1
R	3-phospho-D-glycerate	O=C(O)[C@H](O)COP(=O)(O)O
S	α-D-glucose	OC[C@H]1O[C@H](O)[C@H](O)[C@@H](O)[C@@H]1O
T	α-D-glucose-6-phosphate	O=P(O)(O)OC[C@H]1O[C@H](O)[C@H](O)[C@@H](O)[C@@H]1O
U	triphosphate	O=P(O)(O)OP(=O)(O)OP(=O)(O)O
V	pyrophosphate	O=P(O)(O)OP(=O)(O)O
W	AMP	Nc1ncnc2c1ncn2[C@@H]1O[C@H](COP(=O)(O)O)[C@@H](O)[C@H]1O
X	oxidized [2Fe-2S] ferredoxin	S1[Fe+]S[Fe+]1
Y	reduced [2Fe-2S] ferredoxin	S1[Fe]S[Fe+]1
Z	oxaloacetate	O=C(O)CC(=O)C(=O)O
AA	carbon dioxide	O=C=O
AB	GTP	Nc1nc2c(ncn2[C@@H]2O[C@H](COP(=O)(O)OP(=O)(O)OP(=O)(O)O)[C@@H](O)[C@H]2O)c(=O)[nH]1

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Table S2 (continued from previous page)

Token	Putative biochemical identity	SMILES
AC	GDP	`Nc1nc2c(ncn2[C@@H]2O[C@H](COP(=O)(O)OP(=O)(O)O)[C@@H](O)[C@H]2O)c(=O)[nH]1`
AD	ITP	`O=c1[nH]cnc2c1ncn2[C@@H]1O[C@H](COP(=O)(O)OP(=O)(O)OP(=O)(O)O)[C@@H](O)[C@H]1O`
AE	IDP	`O=c1[nH]cnc2c1ncn2[C@@H]1O[C@H](COP(=O)(O)OP(=O)(O)O)[C@@H](O)[C@H]1O`
AF	thiamine diphosphate (TPP/ThDP)	`Cc1ncc(C[n+]2csc(CCOP(=O)(O)OP(=O)(O)O)c2C)c(N)n1`
AG	2-(α-hydroxyethyl)thiamine diphosphate	`Cc1ncc(C[n+]2c(C(C)O)sc(CCOP(=O)(O)OP(=O)(O)O)c2C)c(N)n1`
AH	coenzyme A	`CC(C)(COP(=O)(O)OP(=O)(O)OC[C@H]1O[C@@H](n2cnc3c(N)ncnc32)[C@H](O)[C@@H]1OP(=O)(O)O)[C@@H](O)C(=O)NCCC(=O)NCCS`
AI	acetyl-CoA	`CC(=O)SCCNC(=O)CCNC(=O)[C@H](O)C(C)(C)COP(=O)(O)OP(=O)(O)OC[C@H]1O[C@@H](n2cnc3c(N)ncnc32)[C@H](O)[C@@H]1OP(=O)(O)O`
AJ	enzyme N^6-(S-acetyldihydrolipoyl)lysine	`*NC(=O)CCCC[C@@H](S)CCSC(C)=O`
AK	enzyme N^6-(dihydrolipoyl)lysine	`*NC(=O)CCCC[C@@H](S)CCS`
AL	enzyme N^6-(lipoyl)lysine	`*NC(=O)CCCC[C@@H]1CCSS1`
AM	oxidized flavodoxin	`CC2(C=C1(N=C3(C(=O)NC(=O)N=C(N(CC(O)C(O)C(O)COP([O-])(=O)[O-])C1=CC(C)=2)3)))`
AN	reduced flavodoxin	`CC2(C=C1(NC3(C(=O)NC(=O)NC(N(CC(O)C(O)C(O)COP([O-])(=O)[O-])C1=CC(C)=2)=3)))`

2.1.2 Automorphisms of the glycolysis reaction graph

Structural automorphism analysis of the species-reaction graph gives group order two. The paired orbits exchange GTP with ITP, GDP with IDP, and reactions r17 with r19. This is an equivalence of the unlabeled reaction structure, not chemical interchangeability of the nucleotide species.

2.1.3 Stoichiometric rank, conservation relations, and Jacobian constraints

The `Props` module analyzes the network structure. For $|X| = 40$ species and $|\mathscr{R}| = 34$ reactions, the stoichiometric matrix $\mathbf{S} \in \mathbb{Z}^{40\times 34}$ yields:

$$\operatorname{rank}(\mathbf{S}) = 23, \qquad \dim\ker(\mathbf{S}^T) = 17, \qquad \dim\ker(\mathbf{S}) = 11. \tag{1}$$

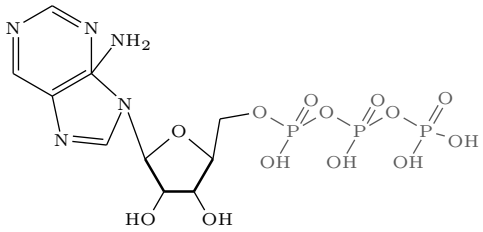
The nullities count independent linear conservation relations and balanced reaction directions. A right-kernel basis need not be non-negative or realizable.

The listed model is conservative and inconsistent, with both verdicts verified by exact rational certificates. A positive conservation vector bounds species inventories, while no all-positive balanced flux involves every reaction. Balanced reaction cycles alone do not establish energy dissipation or thermodynamic infeasibility.

For any differentiable kinetic realization, $\operatorname{rank}(J) \leq \operatorname{rank}(\mathbf{S}) = 23$. A sparsity-pattern matching result can exceed this bound because it ignores algebraic dependencies imposed by $J = \mathbf{S}\,\partial v/\partial c$. Full-space Jacobian singularity therefore does not rule out isolated or stable equilibria within a stoichiometric compatibility class. We make no multistability claim from this check.

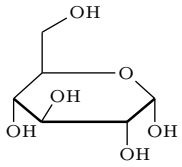
2.1.4 Qualitative reachability of ATP- and ADP-dependent glycolytic routes

(a)



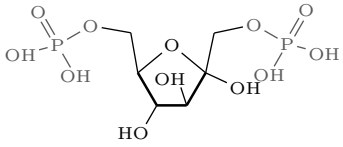
two phosphoanhydride bonds; $r10$ leaves ADP

(b)



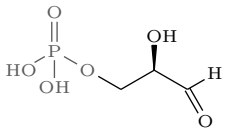
anomeric OH down; no phosphate before $r10$

(c)



cleaved by aldolase $r7$; C2 configuration not specified

(d)



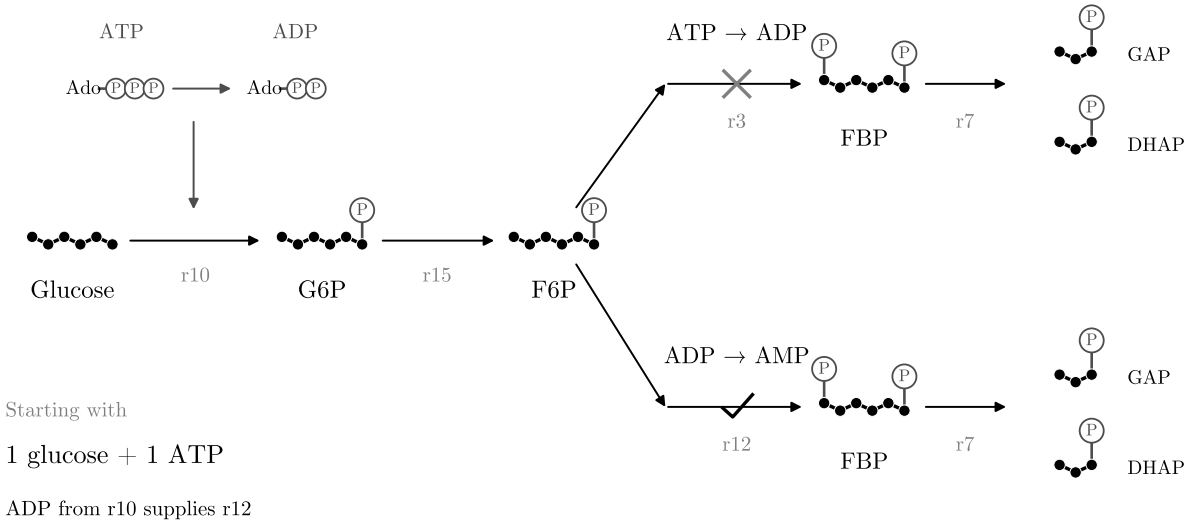
(R) centre; released with DHAP by $r7$

Fig. S2: Key metabolite structures in the extended glycolysis network, drawn from the SMILES of Table S2. **(a)** ATP (C). **(b)** α-D-glucose (S). **(c)** D-fructose 1,6-bisphosphate (H). **(d)** D-glyceraldehyde 3-phosphate (J). Parenthesized tokens match Table S2. Sugars are Haworth projections (bold front bonds, substituents up or down), hydrogens on carbon are implicit, and phosphoryl groups are drawn in gray as in Figure S3. The anomeric center of (C) is drawn without configuration because the listed SMILES leaves it unspecified.

The target is glyceraldehyde-3-phosphate (J), starting from glucose (S) and ATP (C). Both routes pass through fructose-1,6-bisphosphate (H). Figure S2 shows selected metabolite structures. The ADP-dependent phosphorylation in $r12$ corresponds to the reaction recorded as `KEGG R05805` [1].

The two candidate routes each contain four reactions. The ATP-dependent route uses $r10, r15, r3, r7$, whereas the ADP-dependent route replaces $r3$ with $r12$ (Figure S3). Qualitative search admits both because it retains a species in the reachable set after its first use. It therefore does not account for the two ATP molecules consumed by the ATP-dependent sequence.

(a)



(b)

Initial inventory	*ATP-dependent*	*ADP-dependent*
1 glucose + 1 ATP	✗ Blocked	✓ 4 steps
1 glucose + 2 ATP	✓ 4 steps	✓ 4 steps
1 glucose + 1 ATP + 1 F6P	✗ Blocked	✓ 4 steps

Fig. S3: Phosphoryl-donor selectivity in ATP- versus ADP-dependent upper glycolysis. **(a)** Alternative phosphofructokinase steps: $r3$ (ATP-dependent) versus $r12$ (ADP-dependent) following glucose phosphorylation ($r10$) and isomerization ($r15$) to produce FBP, DHAP, and GAP (filled circles represent carbon atoms, circled P represents phosphate, and Ado represents adenosine). The mark on each branch (tick or cross) gives realizability from $1 \times \text{Glc} + 1 \times \text{ATP}$. **(b)** Realizability under cofactor supplementation. The ADP-dependent route is realizable by recycling ADP formed in $r10$, whereas the ATP-dependent route requires a second ATP token (ticks show verified certificates, and crosses indicate exhaustive failure).

2.1.5 Glycolytic pathway realizability under ATP limitation

We checked both candidate routes under discrete marking semantics using `PathwayRealizability`. For an initial marking of exactly one ATP and one glucose token:

$$m_0 = \{\text{ATP} : 1,\ \alpha\text{-D-glucose} : 1\} = \{C : 1, S : 1\}, \tag{2}$$

only the ADP-dependent route ($r10 \to r15 \to r12 \to r7$) has a firing certificate.

In the listed model, hexokinase ($r10$) consumes the available ATP and produces ADP and glucose 6-phosphate. After isomerization ($r15$), the ATP-dependent phosphorylation ($r3$) lacks its donor. The ADP-dependent reaction ($r12$) can instead consume the ADP formed in $r10$, yielding AMP and FBP. Aldolase ($r7$) then produces the two triose phosphates. All four steps of this sequence pass the independent inventory check.

Both routes were evaluated from the three inventories in Figure S3b (one glucose with one ATP, with two ATP, or with one ATP and one F6P). All positive sequences were independently replayed, and all negative results followed exhaustive searches for the prescribed reaction multiset.

Adding F6P does not make the prescribed ATP-dependent sequence realizable, because that sequence still consumes ATP in both $r10$ and $r3$. Supplying a second ATP enables it. This conclusion concerns the specified four reactions, rather than all possible routes in the larger network. It illustrates why a list of reachable metabolites cannot determine whether a finite phosphoryl-donor inventory suffices.

2.2 Precursor requirements in a rule-generated formose network

The formose example uses a finite rule-generated network to compare reachable species sets with realizable reaction sequences.

Four rules for aldol addition, retro-aldol cleavage, and reversible keto-enol tautomerization expand formaldehyde and glycolaldehyde seeds. The fourth iteration contains 37 species and 44 reactions, where iteration is a graph-expansion depth rather than kinetic time.

The targets are an aldotetrose (T_4) and an aldohexose (T_6). Figure S4 shows their constitutional structures and the selected reactions that form them. Stereochemistry is not assigned in this example.

2.2.1 Network expansion under carbohydrate transformation rules

Table S3 reports the recorded formose counts and generation times. The fourth iteration includes 31 species containing at least four carbons, and reaches ten carbons.

Table S3: Network size and generation time across four iterative formose rule-expansion steps from formaldehyde and glycolaldehyde.

Iteration	Species	Reactions	Generation time (s)
1	3	1	0.015
2	5	4	0.002
3	9	10	0.004
4	37	44	0.014

2.2.2 Hydrogen models in rule expansion

The native expander supports two hydrogen models. With explicit hydrogen vertices, hydrogen atoms are added to every molecule, the rule's mapped `[H]` atoms are matched as vertices, and the hydrogen counts of product atoms are re-derived from default valences. With hydrogen counts, each heavy atom carries its hydrogen count as a label. A mapped hydrogen that the rule moves from atom u to atom v becomes the count changes $\Delta h_u = -1$ and $\Delta h_v = +1$, a donor atom must carry at least as many hydrogens as the rule maps on it, and product counts are the reactant counts plus these changes. A product is accepted only when every atom whose count changed has a non-negative even number of non-bonding electrons after its radical electrons are removed, that is, an integral number of lone pairs. Rules with unmapped, created, released, or deleted hydrogens, degree-sensitive queries, or recursive `SMARTS` run with explicit vertices.

Table S4 compares the models. The formose rules all compile to hydrogen counts and generate identical species and reaction sets at every depth. The three controls test hydrogen bookkeeping. Hydrogen-atom abstraction (`[C:1][H:2].[O:3]>>[C:1].[O:3][H:2]`) should give an ethyl radical and water. Proton transfer between ammonium ions and acetate involves bracket atoms with fixed hydrogen counts. The alkene insertion rule (`[C:1]=[C:2].[Pd:3]>>[C:1]([Pd:3])[C:2]`) is deliberately incomplete, because one carbon loses a bond without gaining one. Atom (including hydrogen) and charge balance were checked for every generated reaction. The checks concern conservation, not chemical plausibility of the rules or their products.

Table S4: Explicit hydrogen vertices versus hydrogen counts in native rule expansion. Upper block: formose networks, with expansion times measured in one session after a warm-up run. Lower block: hydrogen-transfer controls, with the number of generated reactions that conserve atoms and charge.

		Hydrogen vertices		Hydrogen counts		
Network	Species	Reactions	Time (s)	Reactions	Time (s)	Identical
Formose, 4 iterations	37	44	0.017	44	0.009	yes
Formose, 5 iterations	302	371	0.176	371	0.109	yes
Formose, 6 iterations	10 572	11 239	9.032	11 239	5.875	yes
Control	Seeds	Balanced / reactions		Balanced / reactions		
H-atom transfer	`CC`, `[OH]`	0 / 1		1 / 1		
Proton transfer	`C[NH3+]`, `CC(=O)[O-]`, `[NH4+]`	0 / 0		4 / 4		
Incomplete insertion rule	`C=CC=O`, `[Pd]`	0 / 12		18 / 18		

2.2.3 Automorphisms of the generated formose network

The nested iteration-four network has 81 bipartite nodes and 122 edges. Structural automorphism analysis reports group order 512 and 18 vertex orbits of size two. These symmetries refer to reaction incidence and the selected graph attributes.

Repeated rule application can create equivalent local reaction environments. Graph symmetry does not establish equal reaction rates, equal concentrations, or chemical equivalence of the associated molecules.

2.2.4 Conservation laws, semiflows, and siphon coverage

The stoichiometric matrix $\mathbf{S} \in \mathbb{Z}^{37\times 44}$ has rank $s = 36$, yielding the nullspace dimensions:

$$\dim\ker(\mathbf{S}^T) = 1, \qquad \dim\ker(\mathbf{S}) = 8. \tag{3}$$

The positive left-kernel vector bounds the non-negative compatibility class. Its primitive coefficients match the recorded molecular carbon counts.

The network is conservative but inconsistent: exact certificates verify a positive conservation vector and exclude a strictly positive balanced flux covering every reaction. This describes the finite generated network, as generation from seeds does not itself introduce inflow or outflow reactions.

Exact enumeration returns one minimal P-semiflow and seven minimal T-semiflows. There are 28 minimal traps and two minimal siphons. The 28 singleton traps were checked directly, and all subsets of the nine remaining species were exhausted to exclude additional minimal traps. Neither siphon contains the

support of a P-semiflow, so the sufficient persistence test does not certify this model. This does not prove that replenishment is necessary or that extinction occurs.

2.2.5 Qualitative reachability of tetrose and hexose targets

Both targets occur in the iteration-four network, including six constitutional $C_6H_{12}O_6$ isomers. Their presence records products generated by the selected rules.

Under qualitative set semantics (`PathwayFinder`), both targets are reachable from the minimal seed pool $M_{\text{min}} = \{\texttt{C=O}, \texttt{O=CCO}\}$ (Table S5):

- T_4 aldotetrose is reachable at depth 2 via glycolaldehyde enolization followed by aldol coupling with parent glycolaldehyde (Figure S4a, b).
- T_6 aldohexose is reachable at depth 3 via enolization followed by two successive aldol additions (Figure S4a–c).

Table S5: Qualitative reachability of aldotetrose (T_4) and aldohexose (T_6) targets from unquantified seeds $M_{\text{min}} = \{\text{formaldehyde}, \text{glycolaldehyde}\}$. Depth indicates the shortest reaction sequence under set reachability.

Target	Present?	Shortest qualitative depth	Reaction sequence
T_4 (aldotetrose)	Yes	2	Tautomerization → aldol addition
T_6 (aldohexose)	Yes	3	Tautomerization → aldol addition → aldol addition

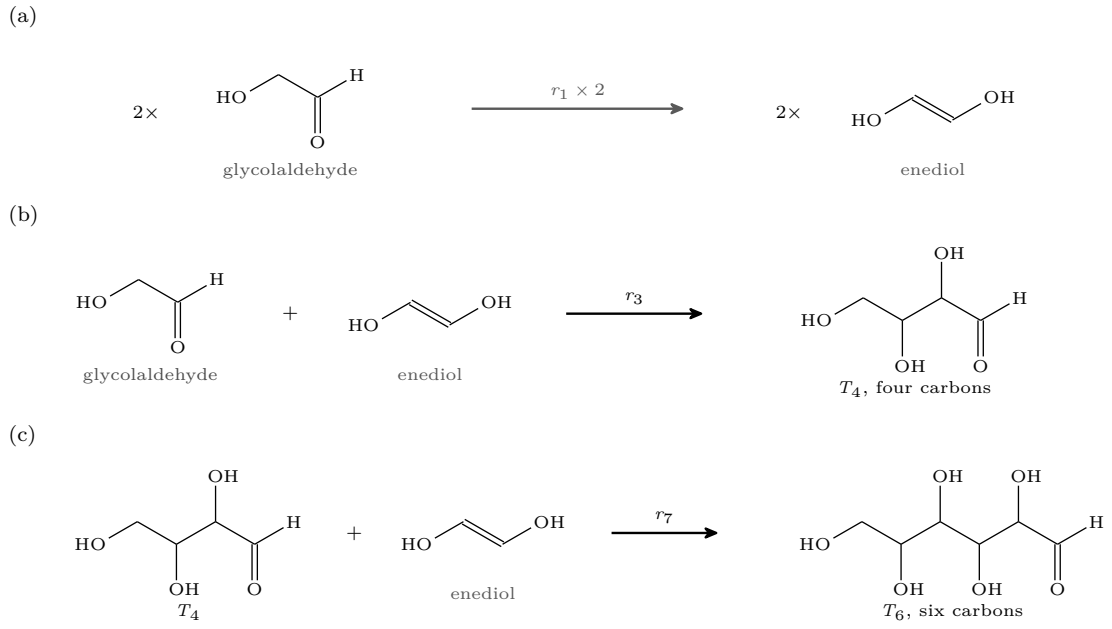


Fig. S4: Stoichiometry and precursor requirements of the selected formose routes. **(a)** Tautomerization (r_1) of glycolaldehyde to enediol. **(b)** Aldol coupling (r_3) to the aldotetrose (T_4). **(c)** Second aldol coupling (r_7) to the aldohexose (T_6). Firing counts are per route. Structures are constitutional: hydrogens on carbon are implicit, and no stereochemistry or atom mapping is asserted.

2.2.6 Precursor-dependent realizability of tetrose and hexose routes

We tested exact discrete realizability across five initial precursor inventories (Table S6): M_1 (minimal seed), M_2 (carbon-exact for C_4, single glycolaldehyde), M_3 (carbon-exact for C_6, formaldehyde-rich), M_4 (two equivalents of each precursor), and M_5 (three glycolaldehyde molecules).

Table S6: Discrete pathway realizability of tetrose (T_4) and hexose (T_6) targets across initial precursor inventories. Total carbon denotes aggregate carbon tokens, Yes indicates a verified firing certificate, while No represents exhaustive failure for the prescribed reaction multiset.

Pool	Initial marking m_0	Total carbon	T_4 realizable?	T_6 realizable?
M_1	{C=O : 1, O=CCO : 1}	3	No	No
M_2	{C=O : 2, O=CCO : 1}	4	No	No
M_3	{C=O : 4, O=CCO : 1}	6	No	No
M_4	{C=O : 2, O=CCO : 2}	6	Yes	No
M_5	{C=O : 0, O=CCO : 3}	6	Yes	Yes

Pools M_4 and M_5 support the selected tetrose route, whereas only M_5 supports the selected hexose route. The tetrose sequence is:

$$\begin{aligned} \texttt{O=CCO} &\longrightarrow \texttt{OC=CO}, \\ \texttt{OC=CO} + \texttt{O=CCO} &\longrightarrow \texttt{O=CC(O)C(O)CO}. \end{aligned} \tag{4}$$

This sequence consumes two glycolaldehyde molecules. One forms the enediol (OC=CO), and the other reacts with that enediol in the aldol-addition step. Pool M_2 contains four carbon atoms but only one glycolaldehyde molecule, so it cannot supply both reactants of the selected route.

The T_4 certificate leaves formaldehyde unchanged. Two glycolaldehyde tokens suffice for this route. This is a route-specific requirement, as a global minimum across alternative pathways would require a separate optimization and completeness argument.

From M_5, the hexose certificate contains four firings: two glycolaldehyde-to-enediol conversions, followed by the tetrose-forming and hexose-forming aldol additions. Three glycolaldehyde molecules suffice for this selected sequence. The qualitative depth of three counts enolization only once, whereas a firing sequence must supply two enediol molecules. These results do not establish minimum seed inventories across unrestricted formose chemistry.

3 Reference networks and structural validation

Table S7 lists the twelve reference networks and the implementation properties they test. The examples target distinct edge cases rather than a random sample of biological networks.

3.1 Algebraic consistency checks and implementation independence

The validation combines expected values, alternative algebraic formulations, checks on returned vectors, and exhaustive small-network enumeration. Some routes share parsing and matrix-construction helpers. For example, complex composition $\mathbf{Y}$ and complex-graph incidence $\mathbf{I}_a$ give $\mathbf{S} = \mathbf{Y}\mathbf{I}_a$, so the deficiency reported from complexes, linkage classes, and stoichiometric rank must agree with a rank difference computed on

Table S7: Reference reaction networks, literature sources, and targeted structural validation properties (invariants reported in main-text Table 1).

Network model	Literature source	Targeted algebraic / topological property
Reversible isomerization	Feinberg [2], Ch. 3	Minimal weakly reversible system, Deficiency Zero Theorem (positive branch)
Irreversible isomerization	Feinberg [2], Ch. 3	Deficiency Zero Theorem (negative branch)
Isomerization cycle	Feinberg [2], Ch. 3	Weak reversibility without individual reversible steps
Three disjoint reversible pairs	Synthetic regression	Linkage-class partitioning on disconnected graphs, where minimal semiflow supports must isolate disjoint pairs rather than their union
Disjoint pairs, non-unit stoichiometry	Synthetic regression	As above, with $E \rightleftharpoons 2F$: exact conservation $2E + F$, which naive coefficient rounding can corrupt
Michaelis-Menten	Michaelis & Menten, Feinberg [2]	Archetypal enzyme mechanism with two conserved moiety pools, deficiency $\delta = 0$, non-weakly reversible
Reversible Michaelis-Menten	Feinberg [2]	Identical deficiency ($\delta = 0$), weak reversibility restored
Edelstein bistable switch	Edelstein [3]	Minimal reversible deficiency-one network admitting multiple steady states
Futile cycle (single-site)	Wang & Sontag [4]	Distributive enzymatic cycle with three independent conservation pools
Futile cycle (two-site)	Wang & Sontag [4]	Multi-stationarity motif isolating failure of numerical left-kernel reconstruction
Shinar-Feinberg ACR motif	Shinar & Feinberg [5]	Absolute concentration robustness, with a minimal siphon that contains no P-semiflow support
Open reactor ($0 \to A \to B \to 0$)	Synthetic benchmark	Explicit zero-complex handling, where cycle closure changes weak reversibility

the complex graph alone:

$$\delta \;=\; n - \ell - \operatorname{rank}(\mathbf{S}) \;=\; \operatorname{rank}(\mathbf{I}_a) - \operatorname{rank}(\mathbf{Y}\mathbf{I}_a). \tag{5}$$

The two sides use different intermediate quantities, so a disagreement localizes the error to the complex graph, the rank routine, or the stoichiometric assembly. Because both sides are evaluated on the same parsed reaction records, the identity checks internal consistency rather than the parse itself, while the external `CRNT4SBML` comparison covers that step with an independent reader.

The checks address two errors: rounding a floating-point kernel basis can violate the exact conservation identity, and inconsistent matrix-row and Petri-place ordering can attach a correct vector to the wrong species. These are reconstruction and indexing errors. They are not evidence that numerical nullspace methods generally fail to approximate the correct subspace.

3.2 Numerical nullspaces and exact integer conservation laws

We compared three operations on the same twelve reference networks: `SciPy`'s floating-point nullspace, the deliberately naive reconstruction round($1000B$) of that basis B, and `SynKit`'s exact integer conservation vectors. Table S8 reports maximum absolute residuals and invalid rounded-vector counts. The scale 1000 is fixed in advance for this illustration, so the result is sensitive to basis choice and is not a comparison against rational-reconstruction algorithms. The zero-dimensional left kernel of the open reactor has no vectors, so its zero residual is vacuous.

The supplementary audit reconstructs the printed 34-reaction network, verifies its species count and rank, checks ATP-dependent and ADP-dependent routes from one and two ATP tokens, and independently

Table S8: Conservation residuals of numerical versus exact integer kernel vectors across the reference networks. SVD residual reports $\max|\mathbf{S}^T B|$ for floating-point basis B, Rounded invalid counts columns of round($1000B$) with non-zero integer residuals, and Exact residual evaluates `SynKit`'s integer basis. The open reactor has a zero-dimensional conservation space.

Network	SVD residual	Rounded invalid	Exact residual
Reversible isomerization	1.1e-16	0	0
Irreversible isomerization	0.0e+00	0	0
Isomerization cycle	2.2e-16	0	0
Disjoint pairs	1.1e-16	0	0
Non-unit disjoint pairs	1.1e-16	0	0
Michaelis-Menten	2.2e-16	1	0
Reversible Michaelis-Menten	1.7e-16	1	0
Edelstein	0.0e+00	0	0
One-site futile cycle	3.3e-16	3	0
Two-site futile cycle	5.3e-16	0	0
Shinar-Feinberg	1.1e-16	0	0
Open reactor	0.0e+00	0	0

replays each positive certificate. It also verifies the selected formose reaction multisets for all five printed inventories, including the three-glycolaldehyde positive control. These are prescribed-flux tests, and alternative reaction multisets are outside their negative conclusions. Complete structural automorphism searches recover the reported orders two and 512.

The positive-kernel decisions are accompanied by exact certificates in the evidence. For $A = \mathbf{S}^{\mathsf{T}}$ (conservativity) or $A = \mathbf{S}$ (consistency), a positive certificate is an integer vector $x > 0$ with $Ax = 0$. A negative certificate is a rational vector y with $A^{\mathsf{T}}y \geq 0$ and at least one positive component: such a vector excludes $x > 0$ because $y^{\mathsf{T}}Ax = (A^{\mathsf{T}}y)^{\mathsf{T}}x > 0$ would contradict $Ax = 0$. Exact certificate verification confirms all four printed decisions. The recorded matrices and witnesses permit verification without a numerical solver.

4 Computational scaling across network families

The C++ main sweep measures directed chains, reversible chains, and random sparse bimolecular networks at four size parameters, with one measurement per configuration and a 300-second per-task limit. The generator also supports cycles, but cycles are not part of that reported sweep. A separate 200 000-evaluation limit applies to siphon enumeration. Each search state and each closure minimization call counts as one evaluation. All 60 requested measurements complete within the recorded limits. No configuration is skipped.

A paired C++ study measures implication-pruned search and exhaustive subset enumeration at 11, 15, 19, and 21 species, with five repeats and alternating order. The evidence stores raw times, medians, and quartiles, and checks equality of every returned siphon set after aligning species identifiers. It supplies the main scaling panel without extrapolated timings.

In the random sparse family, conserved-pool enumeration completes at 395 species in 0.0607 s. Table S9 reports the complete minimal-siphon counts and timings. Their non-monotonic counts and runtimes reflect changes in topology across these fixed seeded instances. These observations depend on topology, software, and hardware and establish neither a polynomial bound for enumeration nor a universal network-size limit.

Table S9: Complete C++ minimal-siphon enumeration for the four seeded random-sparse instances. Each row is a single run; time includes the Python interface.

Species	Reactions	Minimal siphons	Time (s)
49	75	331	0.0474
99	150	272	0.122
197	300	47	0.0147
395	600	87	0.161

5 Supplementary structural analyses and runtime comparisons

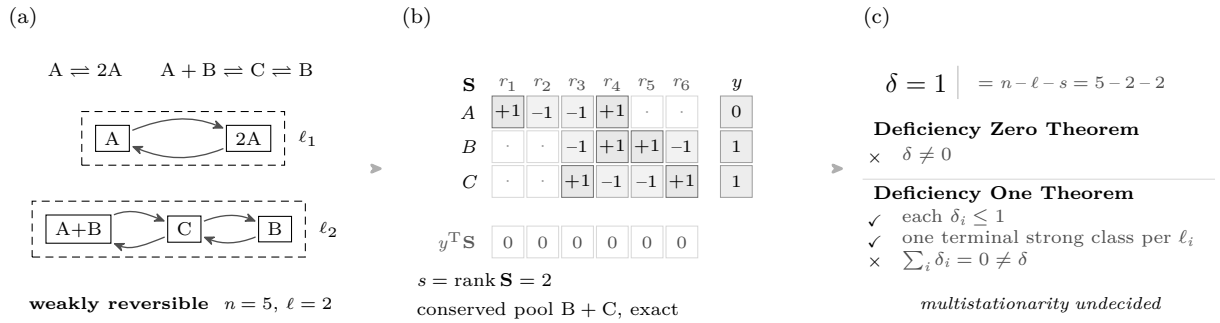


Fig. S5: Conservation structure and deficiency-theorem checks for the Edelstein network [3]. **(a)** Complex graph and its two linkage classes. **(b)** Stoichiometric matrix beside the primitive conservation vector y, whose product $y^{\mathsf{T}}\mathbf{S}$ vanishes over $\mathbb{Q}$ and certifies the conserved pool $[\mathrm{B}] + [\mathrm{C}]$. **(c)** Each classical deficiency theorem fails one hypothesis, leaving multistationarity undecided by deficiency alone.

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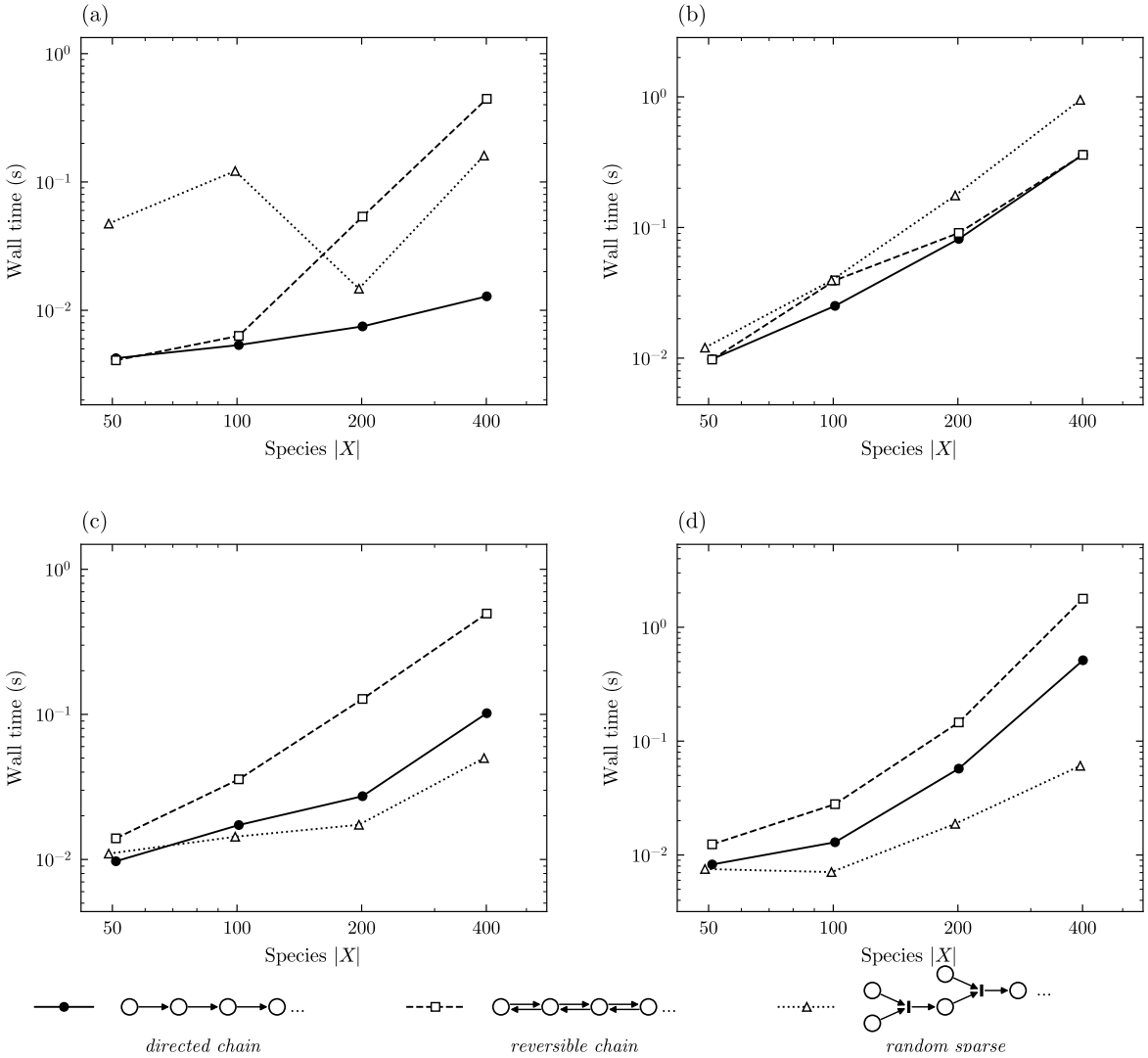


Fig. S6: C++ structural-analysis runtimes across directed-chain, reversible-chain, and random-sparse networks. **(a)** Minimal-siphon enumeration. **(b)** Exact CRNT analysis. **(c)** Graph canonicalization. **(d)** Exact conserved-pool enumeration. Points and connecting lines show completed single-run measurements. The key draws a small member of each family (circles: species, bars: bimolecular reactions) beside its line style. All 60 measurements complete.

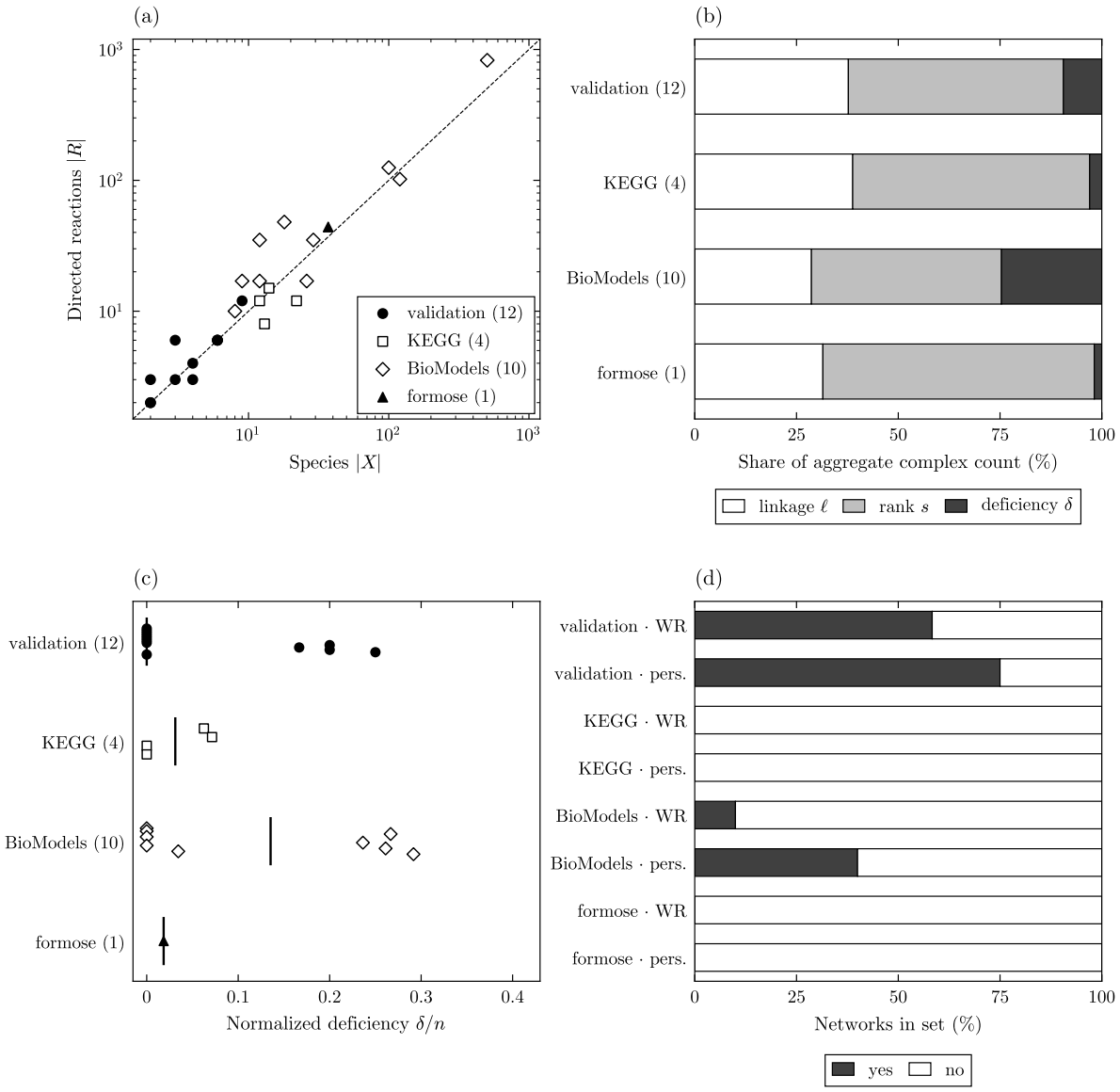


Fig. S7: Structural comparison of reference, `KEGG`, `BioModels`, and formose networks. **(a)** Directed reactions versus species (dashed line: $y = x$). **(b)** Complex-count decomposition $n = \ell + s + \delta$ (ℓ linkage classes, s rank, δ deficiency) by network set. **(c)** Normalized deficiency δ/n (ticks: set medians). **(d)** Weak reversibility (WR) and sufficient siphon persistence coverage (pers.; assessed for every reaction-bearing model). Reaction-free `SBML-qual` models are omitted.